Tip: the journey bar is clickable - jump back to any finished stage, or between your tests, anytime.

RESULTS

Results

01 · MULTIPLE LINEAR REGRESSION

one numeric outcome, several predictors

Table 1. Regression coefficients

	(1)	β	VIF
(Intercept)	41.21		
	(2.33)		
	[36.62, 45.81]		
pretest_score	0.31	—	1.01
	(0.04)		
	[0.22, 0.39]		
study_hours_per_week	1.30	—	1.02
	(0.15)		
	[1.01, 1.60]		
teaching_method: Flipped	5.97	—	1.00
	(0.88)		
	[4.24, 7.71]		
teaching_method: Lecture	−4.68	—	1.00
	(0.88)		
	[−6.41, −2.95]		
Num.Obs.	240		
R^2	0.55		
R^2 Adj.	0.54		
F	70.38		
RMSE	5.49		
AIC	1510.28		
BIC	1531.16		
Log.Lik.	−749.14		

assumption checks: linearity, normality, homoscedasticity, and multicollinearity (VIF).

Figure. Residual diagnostics

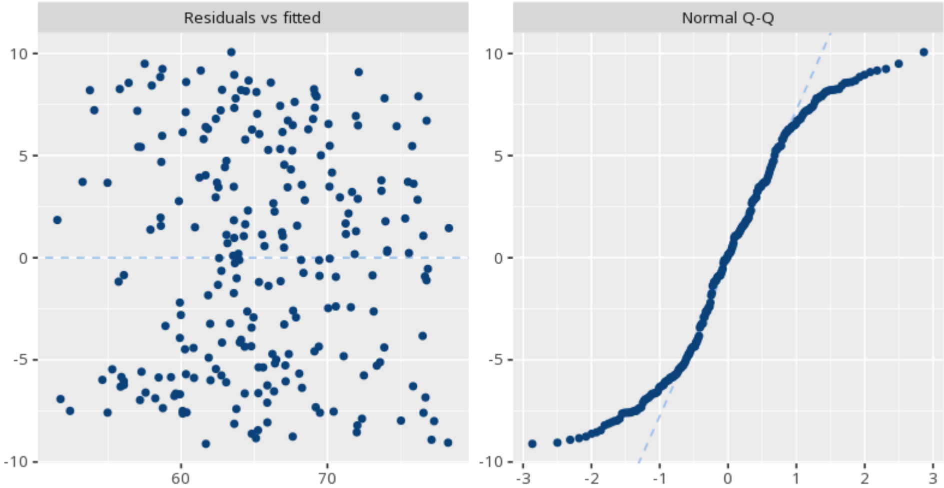
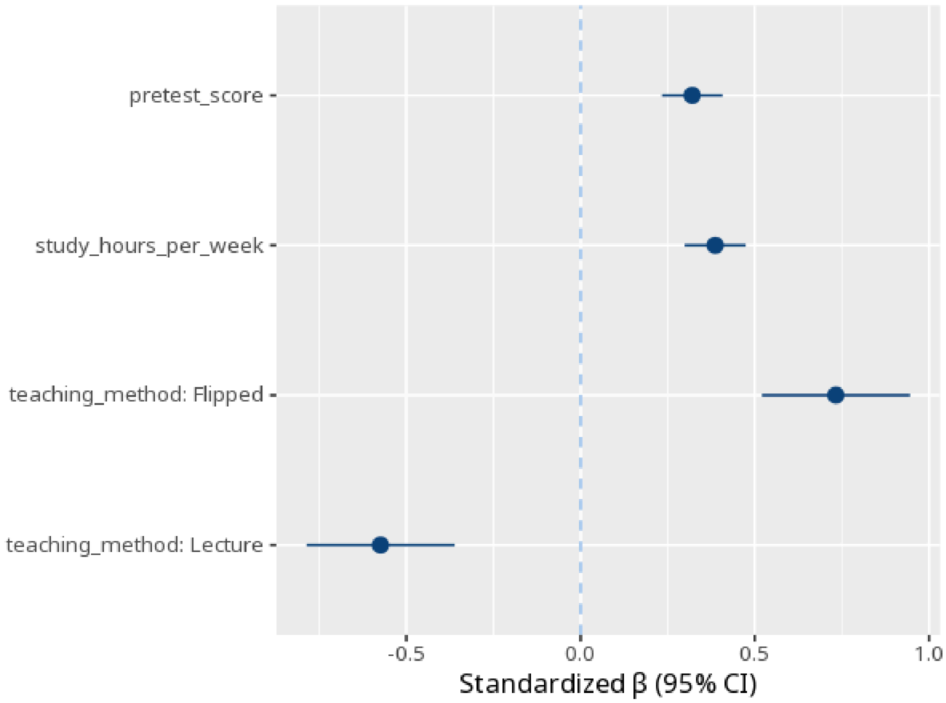


Figure. Coefficient plot



Understanding the numbers

R^2. The share of the outcome's variance jointly explained by all predictors. Here, R^2 = .55, so the model accounts for about 55% of the variance in the outcome.

VIF. How much a predictor's variance is inflated by overlap with the other predictors (multicollinearity); values above ~5-10 are cause for concern (O'Brien, 2007). Here, the largest VIF in this run is 1.02.

RMSE. The root-mean-square error: the typical size of the model's prediction error, in the outcome's own units. Here, RMSE = 5.49.

R^2 Adj.. R^2, adjusted for the number of predictors in the model. Here, adjusted R^2 = 0.54.

AIC. A fit measure that penalizes model complexity, useful for comparing models fit to the same data (lower is better). Here, AIC = 1510.28.

BIC. Like AIC but penalizes complexity more heavily as sample size grows. Here, BIC = 1531.16.

Log.Lik.. The log-likelihood: how probable the observed data are under the fitted model; it underlies AIC and BIC. Here, Log.Lik. = −749.14.

N. The number of observations the model was fit on. Here, N = 240.

F. The overall F-test of whether the model explains more variance than a model with no predictors at all. Here, F = 70.38.

B. Each predictor's effect holding the others constant, in that predictor's own units. Here, predictor pretest_score gave B = 0.31.

How to read this test

Each predictor's B is its effect holding the others constant; p and CI assess it. R^2 is the joint variance explained; VIF flags predictors that overlap too much (multicollinearity). B is in each predictor's own units, so to compare which predictors matter most use the standardized coefficient (β), not B. Your significance threshold (α) is 0.05.

APA template: The model explained R^2=.55 of the variance, F(4,235)=70.38, p < .001; predictor pretest_score gave B=0.31, p < .001.

Statistical basis: Ordinary least squares regression coefficients (B), SE, CI (R Core Team (2026). R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna.); Standardized coefficients (β) (Lüdecke D, Ben-Shachar MS, Patil I, Makowski D (2020). "Extracting, Computing and Exploring the Parameters of Statistical Models using R." Journal of Open Source Software, 5(53), 2445.); Variance inflation factor (VIF) cutoff guidance (O'Brien, R. M. (2007). "A caution regarding rules of thumb for variance inflation factors." Quality & Quantity, 41(5), 673-690.). Full references in the exported CITATIONS.txt.